F7AB - Actuarial and Financial Mathematics
Tutorial 6 – Yields and Measuring fund performance

Question 1 You pay a price of £1 000 for an investment that will repay you £120 at the end of each of the next 10 years. Calculate the yield on this investment.
Question 2 You invest £10 000 in a project and receive back £700.87 per annum payable half-yearly in arrears for the next 25 years. Calculate the yield on this investment.
Question 3 You invest £100 000 in a project which returns £20 000 at the end of each of the first, second and third years, and then £40 000 at the end of each of the fourth, fifth and sixth years. Calculate the yield on this project.
Question 4 You invest in a mining project which requires you to pay £1 000 000 at the start of each of the calendar years 2018, 2019 and 2020 to develop the mine. It is expected to be in active production for twenty years, returning a profit of £250,000 at the end of each calendar year 2024 to 2043 inclusive. Calculate the yield on this project.
Question 5 Assume the set-up of Question 4 of Tutorial 5.

1. In each case considered in Q4.1, write down the equation of value at the end of the loan period and then verify that the yield on the transaction is still 6.5% per annum.

2. In the case considered in Q4.3, write down the equation of value at the end of the loan period and then calculate the yield on the transaction.

Question 6 An investor is considering investing in two possible projects.

- Project A is a smartphone factory, which will cost £200 million immediately to build, and at the end of each of the first, second, third, fourth and fifth years will make a profit of £60 million. At the end of the fifth year the current generation of mobile telephones will be obsolete, and the factory will be sold for £20 million.

- Project B is identical to Project A until the end of the fifth year. At that time the factory will make a profit of £60 million as before, but instead of being sold the factory will be re-tooled at a cost of £180 million to produce the next generation of smartphones. It will then make a profit of £60 million at the end of the sixth, seventh, eighth, ninth and tenth years, and at the end of the tenth year it will be sold for £20 million.

1. Calculate the yield on Project A.

2. Calculate a yield on Project B.

3. State sufficient conditions, involving the signs of the net cash-flows under a financial transaction, under which there exists a unique yield $i^\star \geq -1$ on that transaction.

4. Verify that Project A satisfies the conditions above, but that Project B does not.

5. Prove that the yield on Project B found in part (2) above is unique, despite part (4) above.

Question 7 You have £1 000 000 to invest, and you have a choice of two investments.

- Investment A will return £130 000 at the end of each of the next 10 years, and in addition will return your original investment of £1 000 000 at the end of the 10 years.
- Investment B will return a series of payments, annually in arrears, starting at £160 000 at the end of the first year, then increasing by 5% per annum compound until the final payment is made at the end of the 10th year.

Using the highest yield as the decision-making criterion, determine the investment that you would choose.

Question 8 Suppose you have two series of cashflows:

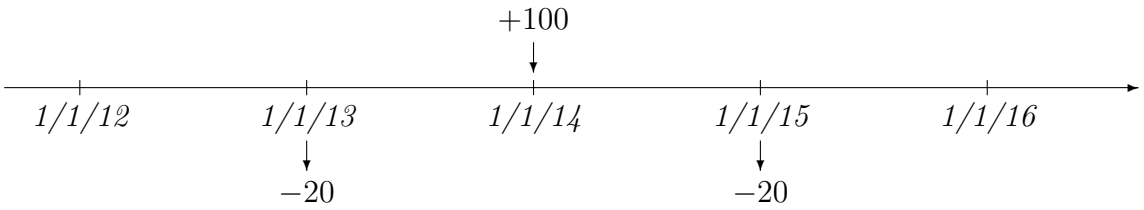
- cashflows of $a_1, a_2, \ldots, a_n$ are payable at times $t_1, t_2, \ldots, t_n$, and
- cashflows of $b_1, b_2, \ldots, b_m$ are payable at times $s_1, s_2, \ldots, s_m$,

for strictly positive integers n and m. Suppose that the yields on both sets of cashflows exist and are equal to the same value $i^\star$.

Show that if you combine the two sets of cashflows into a single transaction, then $i^\star$ is also a solution of the equation of value of the new, combined cashflows.

However, setting $n = m = 2$, find an example to show that it need not be the unique such solution ≥ -1.

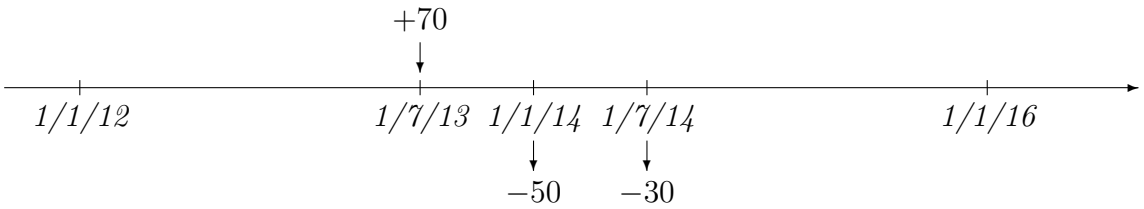
Question 9 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £170.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2014 was £100 just before the cash inflow on 1 Jan 2014 was received, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the two sub-periods calendar years 2012–2013 and 2014–2015.
3. If, in addition, the fund value was £110 on 1 January 2013 and £180 on 1 January 2015 (just before the cashflows on 1 January 2013 and 2015), calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 10 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £140.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.

2. If the fund value on 1 January 2013 was £105, and on 1 January 2015 was £130, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the three sub-periods 2012, 2013–2014 and 2015.

3. Suppose, in addition to the information in part 2, that you knew the following fund values:

Date	1 July 2013	1 January 2014	1 July 2014
Fund Value	£115	£195	£155

   Calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 11 [CT1, Apr 2010, Q10] A pension fund's assets were invested with two fund managers. On 1 January 2007, Manager A was given £120 000 and Manager B was given £100 000. A further £10 000 was invested with each manager on 1 January 2008 and again on 1 January 2009.

The values of the funds were:

Date	31 Dec 2007	31 Dec 2008	31 Dec 2009
Manager A	£130 000	£135 000	£180 000
Manager B	£140 000	£145 000	£150 000

1. Calculate the time-weighted rates of return earned by Manager A and Manager B over the period 1 January 2007 to 31 December 2009.

2. Show that the money-weighted rate of return earned by Manager A over the period 1 January 2007 to 31 December 2009 is approximately 9.4% per annum.

3. Explain, without performing further calculations, whether the money-weighted rate of return earned by Manager B over the period 1 January 2007 to 31 December 2009 was higher than, lower than or equal to that earned by Manager A.

4. Discuss the relative performance of the two fund managers.

Question 12 [CT1, Sep 2011, Q4] A pension fund makes the following investments in £m:

Date	1 Jan 2009	1 Jul 2009	1 Jan 2010
Investment amounts (£m)	1.5	6.0	4.0

The rates of return earned on money invested in the fund were as follows:

Time period	Rate of return (i.e. not annualised)
1 Jan 2009 to 30 Jun 2009	1%
1 Jul 2009 to 31 Dec 2009	2%
1 Jan 2010 to 31 Dec 2010	5%

Assume that 1 January to 30 June and 1 July to 31 December are precise half-year periods.

1. Calculate the time-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

2. Calculate the money-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

Answers: Q1. 0.0345; Q2. 0.0498; Q3. 0.1679; Q4 0.0344; Q5.2. 0.0668; Q6.1 0.1722; Q6.2 0.1722; Q7. B; Q9.1. 1.878%; Q9.2. 3.706%; Q9.3. 3.975%; Q10.1. 9.973%; Q10.2. 9.652%; Q10.3. 9.758%; Q11.1 A: 9.050% p.a. B: 9.409% p.a.; Q11.3. Lower; Q12.1. 4.005% p.a.; Q12.2 4.63% p.a